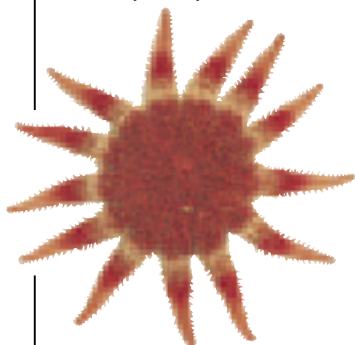


Life in the oceans



Bloody Henry starfish

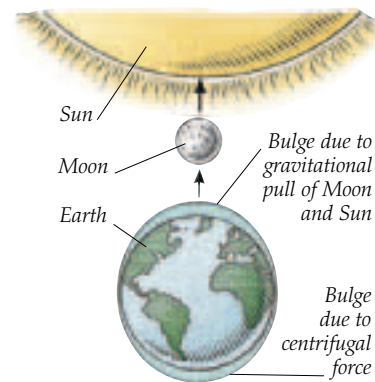


Common sunstar

SHORE LIFE

Often found on the shore at low tide, starfish also live in deeper water. Sea life on the shore must either be tough enough to withstand drying out, or shelter in rock pools. The toughest animals and plants live high on the shore, but the least able to cope in air are found at the bottom.

FROM THE SEA SHORE to the deepest depths, oceans are home to some of the most diverse life on Earth. Animals live either on the seabed or in midwater where they swim or float. Plants are only found in the sunlit zone where there is enough light for them to grow either anchored to the bottom or drifting in the water. Animals are found at all depths of the oceans, but are most abundant in the sunlit zone where food is plentiful. Not all free-swimming animals stay in one zone—the sperm whale dives to over 1,650 ft (500 m) to feed on squid, returning to the surface to breathe air. Some animals from cold, deep waters, such as the Greenland shark in the Atlantic, are also found in the cold, surface waters of polar regions. Over 90 percent of all species dwell on the bottom. One rock can be home to at least 10 major types, such as corals, mollusks, and sponges. Most ocean animals and plants have their origins in the sea, but some like whales and sea grasses are descended from ancestors that once lived on land.



TIME AND TIDE

Anyone spending time by the seaside or in an estuary will notice the tides. Tides are caused by the gravitational pull of the Moon on the Earth's mass of seawater. An equal and opposite bulge of water occurs on the side of the Earth away from the Moon, due to centrifugal force. As the Earth spins on its axis, the bulges (high tides) usually occur twice a day in any one place. The highest and lowest tides occur when the Moon and Sun are in line causing the greatest gravitational pull. These are the spring tides at new and full Moon.

SQUISHY SQUID

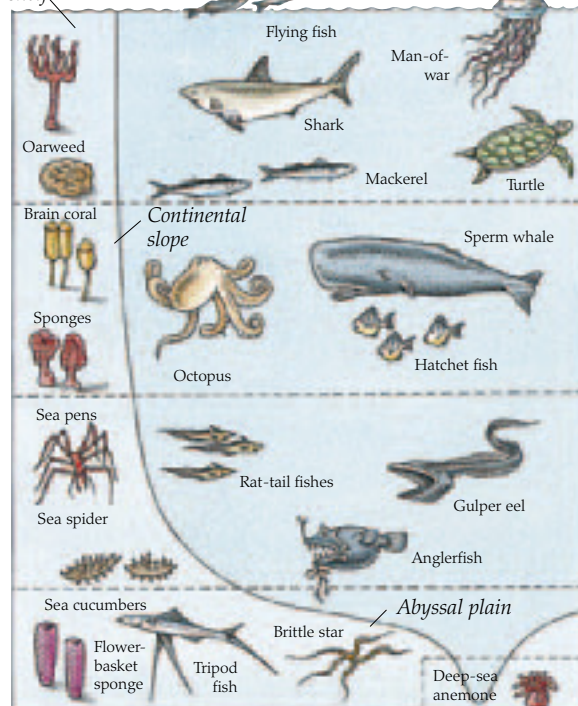
Squid are among the most common animals living in the ocean. Like fish, they often swim around in shoals for protection in numbers. Their torpedo-shaped bodies are streamlined so they can swim fast.

Inside squid's soft body is a horny, penlike shell

Funnel expels jet of water for moving in the sea

Tentacles reach out to grasp food

Continental shelf



Note: Neither the marine life nor zones are drawn to scale

Sunlit zone
0–650 ft
(0–200 m)

Twilight zone
650–3,300 ft
(200–1,000 m)

Dark zone
3,300–13,000 ft
(1,000–4,000 m)

Abyss
13,000–20,000 ft
(4,000–6,000 m)

Trench
Over 20,000 ft
(6,000 m)

Deep-sea cat shark grows to only 20 in (50 cm) long

THE OCEAN'S ZONES

The ocean is divided up into broad zones, according to how far down sunlight penetrates, and water temperature and pressure. In the sunlit zone, there is plenty of light, much water movement, and seasonal changes in temperature. Beneath this is the twilight zone, the maximum depth where light penetrates. Temperatures here decrease rapidly with depth to about 41°F (5°C). Deeper yet is the dark zone, where there is no light and temperatures drop to about 34–36°F (1–2°C). Still in darkness and even deeper is the abyss and then the trenches. There are also zones on the seabed. The shallowest zone lies on the continental shelf. Below this are the continental slope, the abyssal plains, and the seafloor trenches.